

Vaxcel® FLUCONAZOLE 2mg/ml IV INFUSION

COMPOSITION:

Each ml contains:
Fluconazole

2mg

PRESENTATION:

Clear Solution.

INDICATIONS:

Therapy may be instituted before the results of the cultures and other laboratory studies are known; however, once these result become available, anti-infective therapy should be adjusted accordingly.

1. Cryptococcosis, including cryptococcal meningitis and infections of other sites (e.g., pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplants or other causes of immunosuppression may be treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.

2. Systemic candidiasis, including candidemia, disseminated candidiasis and other forms of invasive candidal infection. These include infections of the peritoneum, endocardium, eye and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated.

3. Mucosal candidiasis. These include oropharyngeal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated. Prevention of relapse of oropharyngeal candidiasis in patients with AIDS.

4. Genital candidiasis. Vaginal candidiasis, acute or recurrent; and prophylaxis to reduce the incidence or recurrent vaginal candidiasis (3 or more episodes a year). Candidal balanitis.

5. Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.

6. Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, tinea versicolor, and dermal candida infections.

MECHANISMS OF ACTION:

Fluconazole is a triazole antifungal agent, is a potent and specific inhibitor of fungal sterol synthesis. Both orally and intravenously administered fluconazole was active in a variety of animal fungal infection models. Activity has been demonstrated against opportunistic mycoses, such as infections with *Candida* spp. including systemic candidiasis in immunocompromised animals; with *Cryptococcus*

neoformans, including intracranial infections; with *Microsporum* spp. and *Trichophyton* spp. Fluconazole has also been shown to be active in animal models in endemic mycoses, including infections with *Blastomyces dermatitidis*, with *Coccidioides immitis*, including intracranial infection; and with *Histoplasma capsulatum* in normal and immunosuppressed animals. There have been reports of cases of superinfection with *Candida* species other than *C. albicans*, which are often inherently non susceptible to fluconazole (e.g. *Candida krusei*). Such cases may require alternative antifungal therapy. Fluconazole is highly specific for fungal cytochrome P-450 dependent enzymes. Fluconazole 50mg daily given up to 28 days has been shown not to affect testosterone plasma concentrations in males or steroid concentrations in females of child-bearing age. Fluconazole 200 to 400mg daily has no clinically significant effect on endogenous steroid levels or on ACTH stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50mg do not affect its metabolism.

PHARMACOKINETICS:

The pharmacokinetic properties of fluconazole are similar following administration by the intravenous or oral route. After oral administration fluconazole is well absorbed and plasma levels (and systemic bioavailability) are over 90% of the levels achieved after intravenous administration. Oral absorption is not affected by concomitant food intake. Peak plasma concentrations in the fasting state occur between 0.5 and 1.5 hours post-dose with a plasma elimination half-life of approximately 30 hours. Plasma concentrations are proportional to dose. Ninety percent steady-state levels are reached by day 4-5 with multiple once daily dosing. Administration of a loading dose (on day 1) of twice the usual daily dose enables plasma levels to approximate to 90% steady-state levels by day 2. The apparent volume of distribution approximates to total body water. Plasma protein binding is low (11-12%). Fluconazole achieves good penetration in all body fluids studied. The levels of fluconazole in saliva and sputum are similar to plasma levels. In patients with fungal meningitis, fluconazole levels in the CSF are approximately 80% of the corresponding plasma levels. High skin concentrations of fluconazole, above serum concentrations, are achieved in the stratum corneum, epidermis-dermis and eccrine sweat. Fluconazole accumulates in the stratum corneum. At a dose of 50mg once daily, the concentration of fluconazole after 12 days was 73 microgram/g and 7 days after cessation of treatment the concentration was still 5.8 microgram/g. The major route of excretion is renal, with approximately 80% of the administered dose appearing in the urine as unchanged drug. Fluconazole clearance is proportional to creatinine clearance. There is no evidence of circulating metabolites. The long plasma elimination half-life provides the basis for single dose therapy for genital candidiasis and once daily dosing for other indications. A study compared the saliva and plasma concentrations of a single fluconazole 100mg dose administration in a capsule or in an oral suspension by rinsing and retaining in mouth for 2 minutes and swallowing. The maximum concentration of fluconazole in saliva after the suspension was observed 5 minutes after ingestion, and was 182 times higher than the maximum saliva concentration after the capsule which occurred 4 hours after ingestion. After about 4 hours, the saliva concentrations of fluconazole were similar. The mean AUC (0-96) in saliva was significantly greater after the suspension compared to the capsule. There was no significant difference in the elimination rate from saliva or the plasma pharmacokinetic parameters for the two formulations.

Pharmacokinetics in Children:

In children, the following pharmacokinetic data have been reported:

Age Studied	Dose (mg/kg)	Half-life (hours)	AUC (microgram.h/ml)
11 days - 11 months	Single-IV 3mg/kg	23	110.1
9 months - 13 years	Single-Oral 2mg/kg	25.0	94.7
9 months - 13 years	Single-Oral 8mg/kg	19.5	362.5
5 years - 15 years	Multiple IV 2mg/kg	17.4*	67.4*
5 years - 15 years	Multiple IV 4mg/kg	15.2*	139.1*
5 years - 15 years	Multiple IV 8mg/kg	17.6*	196.7*
Mean Age 7 Years	Multiple Oral 3mg/kg	15.5	41.6

*Denotes final day

In premature new-borns (gestational age around 28 weeks), intravenous administration of fluconazole of 6mg/kg was given every third day for a maximum of five doses while the premature new-borns remained in the intensive care unit. The mean half-life (hours) was 74 (range 44-185) on day 1 which decreased with time to a mean of 53 (range 30-131) on day 7 and 47 (range 27-68) on day 13. The area under the curve (microgram.h/ml) was 271 (range 173-385) on day 1 and increased with a mean of 490 (range 292-734) on day 7 and decreased with a mean of 360 (range 167-566) on day 13. The volume of distribution (ml/kg) was 1183 (range 1070-1470) on day 1 and increased with time to a mean of 1184 (range 510-2130) on day 7 and 1328 (range 1040-1680) on day 13.

DOSAGE AND ADMINISTRATION:

Fluconazole may be administered either orally or by intravenous infusion at a rate of approximately 5-10ml/min, the route being dependent on the clinical state of the patient. On transferring from the intravenous route to the oral route or vice versa, there is no need to change the daily dose. The daily dose of Fluconazole should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single dose therapy. Therapy for those types of infections requiring multiple dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An inadequate period of treatment may lead to recurrence of active infection. Patients with AIDS and cryptococcal meningitis usually require maintenance therapy to prevent relapse.

Use in adults :

1. Mucosal Candidiasis

Oropharyngeal candidiasis - the usual dose is 50mg-100mg once daily for 7 - 14 days. Treatment should not normally exceed 14 days except in severely immunocompromised patients. For atrophic oral candidiasis associated with dentures - the usual dose is 50mg once daily for 14 days administered concurrently with local antiseptic measures to the denture. For other candidal infections of mucosa except genital candidiasis, e.g. oesophagitis, non-invasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis etc., the usual effective dose is 50mg daily, given for 14 - 30 days. In unusually difficult cases of mucosal candidal infections the dose may be increased to 100mg daily.

2. For tinea pedis, corporis, cruris, versicolor and dermal *Candida* infections the recommended dosage is 50mg once daily. Duration of treatment is normally 2 to 4 weeks but tinea pedis may require treatment for up to 6 weeks. Duration of treatment should not exceed 6 weeks.

3. For candidemia, disseminated candidiasis and other invasive candidal infections the usual dose is 400mg on the first day followed by 200mg daily. Depending on the clinical response the dose may be increased to 400mg daily. Duration of treatment is based upon the clinical response.

4. a) For cryptococcal meningitis and cryptococcal infections at other sites, the usual dose is 400mg on the first day followed by 200mg - 400mg once daily. Duration of treatment for cryptococcal infections will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis.

b) For the prevention of relapse of cryptococcal meningitis in patients with AIDS, after the patient receives a full course of primary therapy, Fluconazole may be administered indefinitely at a daily dose of 200mg.

5. For the prevention of fungal infections in immunocompromised patients considered at risk as a consequence of neutropenia following cytotoxic chemotherapy or radiotherapy, the dose should be 50 to 400mg once daily, based on the patient's risk for developing fungal infection. For patients at high risk of systemic infection, e.g. patients who are anticipated to have profound or prolonged neutropenia such as during bone marrow transplantation, the recommended dose is 400mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia, and continue for 7 days after the neutrophil count rises above 1000 cells per mm³.

Use in Children :

As with similar infections in adults, the duration of treatment is based on the clinical and mycological response. The maximum adult daily dosage should not be exceeded in children. Fluconazole is administered as a single dose each day. The recommended dosage of fluconazole for mucosal candidiasis is 3mg/kg daily. A loading dose of 6mg/kg may be used on the 1st day to achieve steady-state levels more rapidly. For the treatment of systemic candidiasis and cryptococcal infection, the recommended dosage is 6-12mg/kg daily, depending on the severity of the disease. For the prevention of fungal infections in immunocompromised patients considered at risk as a consequence of neutropenia following cytotoxic chemotherapy or radiotherapy, the dose should be 3-12mg/kg daily, depending on the extent and duration of the induced neutropenia (see use in adult). (For Children with impaired renal functions, see Use in Renal Impairment).

Use in Children below 4 weeks of age and younger :

Neonates excrete fluconazole slowly. In the first 2 weeks of life, the same mg/kg dosing as in older children should be used but administered every 72 hrs. During weeks 3-4 of life, the same dose should be given every 48 hours.

Use in Elderly :

Where there is no evidence of renal impairment, normal dosage recommendations should be adopted. For patients with renal impairment (creatinine clearance <50ml/min) the dosage schedule should be adjusted as described as follows.

Use in Renal Impairment :

Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single-dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of fluconazole, an initial loading dose of 50-400mg should be

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given. After the loading dose, the daily dose (according to indication) should be based on the following table.

Creatinine Clearance (ml/min)	Dosage Intervals / Daily Dose
> 50	100% or 24 hours
< 50 (no dialysis)	50% or 48 hours
Regular Haemodialysis	100% after each dialysis

These are suggested dose adjustments based on pharmacokinetics following administration of single doses. Further adjustment may be needed depending on clinical condition.

CONTRAINDICATIONS:

Fluconazole should not be used in patients with known sensitivity to the drug, any of the inert ingredients or to related azole compounds. Coadministration of terfenadine is contraindicated in patients receiving fluconazole at multiple doses of 400mg per day or higher based upon results of multiple dose interaction study (see Interaction with Other Medicaments). Coadministration of cisapride is contraindicated in patients receiving fluconazole (see Interaction with Other Medicaments). Fluconazole is generally unadvised in association with halofantrine. Fluconazole should not be administered in the following cases:

- Hypersensitivity to fluconazole and/or other azole derivatives
- Pregnancy and lactation (see Pregnancy and Lactation)
- In combination with cisapride and pimozide (see Interaction with Other Medicaments)

WARNING & PRECAUTIONS:

Fluconazole has been associated with rare cases of serious hepatic toxicity including fatalities, primarily in patients with serious underlying medical conditions. In cases of fluconazole-associated hepatotoxicity, no obvious relationship to total daily dose, duration of therapy, sex or age of patient has been observed. Fluconazole hepatotoxicity has usually been reversible on discontinuation of therapy. Patients who develop abnormal liver function tests during fluconazole therapy should be monitored for the development of more serious hepatic injury. Fluconazole should be discontinued if clinical signs or symptoms consistent with liver disease develop that may be attributable to fluconazole. Patients have rarely developed exfoliative cutaneous reactions such as Stevens-Johnson syndrome and toxic epidermal necrolysis, during treatment with fluconazole. AIDS patients are more prone to the development of severe cutaneous reactions to many drugs. If a rash, which is considered attributable to fluconazole, develops in a patient treated for a superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive/systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop. The co-administration of fluconazole at doses lower than 400mg/day with terfenadine should be carefully monitored. (See Interactions with Other Medicaments). In rare cases, as with other azoles, anaphylaxis has been reported. Some azoles, including fluconazole, have been associated with prolongation of the QT interval of the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT prolongation and torsade de pointes in patients taking fluconazole. These reports included seriously ill patients with multiple confounding risk factors, such as structural heart disease, electrolyte abnormalities and concomitant medications that may have been contributory. Fluconazole should be administered with caution to patients with

these potentially proarrhythmic conditions.

Use During Pregnancy :

There are no adequate and well controlled studies in pregnant women. There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of fluconazole as a single or repeated dose in the first trimester. Use in pregnancy should be avoided except in patients with severe or potential life-threatening fungal infections in whom fluconazole may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus. Effective contraceptive measures should be considered in women of child-bearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose. There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high dose (400 mg/day to 800 mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5 mg/kg or 10 mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 50 mg/kg and higher doses. At doses ranging from 80 mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg embryoethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification. These effects are consistent with the inhibition of estrogen on pregnancy, organogenesis and parturition. Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800 mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

Use During Lactation :

Fluconazole is found in human breast milk at concentrations similar to plasma. Breast-feeding may be maintained after a single dose of 150mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole.

INTERACTION WITH OTHER MEDICAMENTS:

Anticoagulants: In post-marketing experience, as with other azole antifungals, bleeding events (bruising, epistaxis, gastrointestinal bleeding, hematuria and melena) have been reported, in association with increases in prothrombin time in patients receiving fluconazole concurrently with warfarin. Prothrombin time in patients receiving coumarin-type anticoagulants should be carefully monitored.

Azithromycin: There was no significant pharmacokinetic interaction between fluconazole and azithromycin.

Benzodiazepines (Short-Acting): Following oral administration of midazolam, fluconazole resulted in substantial increases in midazolam concentrations and psychomotor effects. This effect on midazolam appears to be more pronounced following oral administration of fluconazole than with fluconazole administered IV. If concomitant benzodiazepine therapy is necessary in patients being treated with fluconazole, consideration should be given to decreasing the benzodiazepine dosage and the patients should be appropriately monitored.

Cisapride: There have been reports of cardiac events, including torsade de pointes, in patients to whom fluconazole and cisapride were

co-administered. Co-administration of cisapride is contraindicated in patients receiving fluconazole (see Contraindications).

Cyclosporin: Cyclosporin plasma concentration monitoring in patients receiving fluconazole is recommended.

Oral Contraceptives: Multiple dose use of fluconazole at these doses is unlikely to have an effect on the efficacy of the combined oralshoudl contraceptive.

Phenytoin: Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to a significant degree. If it is necessary to administer both drugs concomitantly, phenytoin levels should be monitored and the phenytoin dose adjusted to maintain therapeutic levels.

Pimozide: Combination with fluconazole will increase risk of ventricular arrhythmia troubles, notably torsade de pointes.

Rifabutin: Patients receiving rifabutin and fluconazole concomitantly be carefully monitored.

Rifampicin: Concomitant administration of fluconazole and rifampicin resulted in a 25% decrease in the AUC and a 20% shorter half-life of fluconazole. In patients receiving concomitant rifampicin, an increase of the fluconazole dose should be considered.

Sulfonyleureas: Fluconazole has been shown to prolong the serum half-life of concomitantly administered sulfonyleureas (chlorpropamide, glibenclamide, glipizide and tolbutamide) in healthy volunteers. Fluconazole and oral sulfonyleureas may be co-administered to diabetic patients but the possibility of a hypoglycemic episode should be borne in mind.

Tacrolimus: There have been reports that an interaction exists when fluconazole is administered concomitantly with tacrolimus, leading to increased serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were co-administered. Patients receiving tacrolimus and fluconazole concomitantly should be carefully monitored.

Terfenadine: The combined use of fluconazole at doses >400mg with terfenadine is contraindicated (see Contraindications). The co-administration of fluconazole at doses <400mg/day with terfenadine should be carefully monitored.

Theophylline: Patients who are receiving high dose theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and therapy modified appropriately if signs of toxicity develop.

Zidovudine: The AUC of zidovudine significantly increased during co-administration with fluconazole. Patients receiving this combination should be monitored for the development of zidovudine-related adverse reactions.

The use of fluconazole in patients concurrently taking astemizole or other drugs metabolized by the cytochrome P-450 system may be associated with elevations in serum levels of these drugs. In the absence of definitive information, caution should be used when co-administering fluconazole. Patients should be carefully monitored. Interaction studies have shown that when oral fluconazole is co-administered with food, cimetidine, antacids or following total body irradiation for bone marrow transplantation, no clinically significant impairment of fluconazole absorption occurs.

Physicians should be aware that drug-drug interaction studies with other medications have not been conducted, but such interactions may occur.

Incompatibilities:

Fluconazole intravenous infusion is compatible with the following administration fluids.

- a) Dextrose 20%
- b) Ringer's Solution
- c) Hartmann's Solution

- d) Potassium Chloride in Dextrose
- e) Sodium Bicarbonate 4.2%
- f) Aminofusin
- g) Normal Saline

Fluconazole may be infused through an existing line with one of the above listed fluids. Although no specific incompatibilities have been noted, mixing with any other drug prior to infusion is not recommended.

SIDE EFFECTS:

Fluconazole is generally well tolerated. The most common undesirable effects observed during clinical trials and associated with fluconazole are:

Nervous System Disorders: Headache.

Gastrointestinal Disorders: Abdominal pain, diarrhea, flatulence and nausea.

Hepatobiliary Disorders: Hepatic toxicity including rare cases of fatalities, elevated alkaline phosphatase, elevated bilirubin, elevated SGOT and SGPT.

Skin and Subcutaneous Tissue Disorders: Rash.

In some patients, particularly those with serious underlying diseases such as AIDS and cancer, changes in renal and hematological function test results and hepatic abnormalities (see Precautions) have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain.

In addition, the following undesirable effects have occurred during post-marketing:

Blood and Lymphatic System Disorders: Leukopenia including neutropenia and agranulocytosis, thrombocytopenia.

Immune System Disorders: Anaphylaxis (including angioedema, face edema, pruritus and urticaria).

Metabolism and Nutrition Disorders: Hypercholesterolemia, hypertriglyceridemia and hypokalemia.

Nervous System Disorders: Dizziness, seizures and taste perversion.

Cardiac Disorders: QT prolongation and torsade de pointes (see Warning & Precautions).

Gastrointestinal Disorders: Dyspepsia and vomiting.

Hepatobiliary Disorders: Hepatic failure, hepatitis, hepatocellular necrosis and jaundice.

Skin and Subcutaneous Tissue Disorders: Alopecia, exfoliative skin disorders including Stevens-Johnson syndrome and toxic epidermal necrolysis.

SYMPTOMS AND TREATMENT OF OVERDOSE:

In the event of overdosage, symptomatic treatment may be adequate. Fluconazole is mostly excreted in urine; force volume diuresis would probably increase the elimination rate. A three hour hemodialysis session decrease plasma levels by approximately 50%.

STORAGE:

Store below 30°C. Do not freeze. For single use only. Discard any remaining solution.

PACK QUANTITIES:

Available in 50ml per infusion bottle.



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